

minute information on infectious disease threats or outbreaks. In the energy sector, Big Data helps oil and gas companies identify potential drilling locations and monitor pipeline operations; similarly, utilities use it to monitor electrical grids.

Financial and insurance companies use Big Data for risk management and real-time analysis of market data. Manufacturers and logistics companies rely on it to manage their supply chains and optimise delivery routes. Other government uses include emergency response, crime prevention, and smart city initiatives.

Comparative analysis: This includes examining user behaviour metrics and observing customer engagement in real-time to compare a company's products, services, and brand awareness with those of its competitors.

Listening to social networks: This is information about what people are saying on social media about a specific company or product that goes beyond what can be delivered in a survey. This data can be used to help identify target audiences for marketing campaigns by observing activity related to specific topics from various sources.

Marketing analysis: This information can be used to make the promotion of new products, services and initiatives more informed and innovative.

Customer satisfaction and feelings: All the information collected can reveal what customers think of a company or brand, how brand loyalty can be preserved if potential problems arise, and how customer service efforts can be improved.

Open source tools for Big Data analytics

Apache Hadoop

This allows distributed processing of large data sets across clusters of computers. It is one of the best Big Data tools designed to scale up from single servers to thousands of machines.

Features

- The cluster is highly scalable and fault tolerant.

- It is based on the 'Data Locality' concept.
 - It allows for faster data processing.
- URL: <https://hadoop.apache.org/releases.html>

Cassandra

The Apache Cassandra database is widely used nowadays for effective management of large amounts of data.

Features

- Supports multi data centre replication. Data is automatically replicated to multiple nodes.
- Most suitable for companies that can't afford to lose data, even when an entire data centre is down.

URL: <http://cassandra.apache.org/download/>

Hive

Hive allows programmers to analyse huge data sets on Hadoop. It helps with querying and managing large data sets quickly.

Features

- It supports SQL like query language (called HQL – Hive Query Language) for interaction and data modelling.
- It supports partition, Bucket and tables.
- Hive is designed for managing and querying only structured data.
- It offers a Java database connectivity (JDBC) interface.

URL: <https://hive.apache.org/downloads.html>

Challenges

- **Lack of understanding of Big Data:** Companies fail in their Big Data initiatives because of a lack of understanding. Employees may not know what the data is, how it is stored, how it is processed, and where it comes from. While data professionals may know what is going on, the others may not have a transparent picture.
- **Confusion in choosing which Big Data tool to use:** Organisations are often confused when choosing the simplest tool for analysing and storing huge data. Is HBase or Cassandra the easiest data storage technology to use? Is Hadoop MapReduce okay, or is Spark a much better option for data analysis and storage? These are the questions that annoy companies, and sometimes they can't find the answers. So they end up making the wrong decisions and choosing the wrong technology. This wastes money, time, effort, and man-hours.
- **Data security:** Securing this huge amount of data is a big challenge too. Companies are often so busy comprehending, storing, and analysing their data sets that they postpone action on data security, becoming victims of malicious hackers.

If you want to help your business achieve more, leveraging Big Data and AI is a must today or there is a clear danger of the organisation falling by the wayside. **END** 

References

- [1] <https://www.statista.com/statistics/871513/worldwide-data-created/>
- [2] <https://www.forbes.com/sites/rkulkarni/2019/02/07/big-data-goes-big/?sh=5b985d0920d7>

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